

Basic Concepts of Economics

Description of Economics: The word **Economics** comes from Ancient Greek: **Oikos** = house **Nomos** = management. At the smallest level, economics means **managing a household**, similar to how a mother manages monthly expenses, buys food, groceries, hires help, etc., all within limited income. This household-level management can be expanded to larger scales: Town, City, Province, National, Regional, Global level

Economics is a social science that studies: **Limited/scarc resources, Unlimited human wants**

How individuals, households, businesses, and governments make choices to allocate resources like: Money, Materials, Manpower, Machinery

Definitions of Economics

1. **Alfred Marshall (19th century):** Economics is a study of mankind in the ordinary business of life. Everyday decisions like buying clothes, choosing food, or finding a job involve: Scarcity, Cost, Choices between alternatives
2. **Economics studies how people make choices** to get what they want under scarcity. Known as the study of scarcity and choice.
3. **Economics focuses on decisions and outcomes related to:** Production, Distribution, Consumption, Goods, Services, Investment, Income
4. **Economic studies cover:** Households, Businesses, Governments, Foreign countries

Economic Theories: Economic theories began to develop **after the Industrial Revolution**. Over the last three centuries, three major economists shaped economic thought **1. Adam Smith 2. Karl Marx 3. John Maynard Keynes**

- All three used common economic elements: Production, Distribution, Choice, Scarcity, Alternate uses

Microeconomics: **Micro** = small (from Greek *mikro-*) Microeconomics studies: Individuals and firms. How they decide to allocate limited resources. How they interact in markets

- **Goals & Focus of Microeconomics:** Analyze market mechanisms that determine: Prices of goods and services, Allocation of scarce resources. Show conditions where free markets lead to desirable outcomes. Study market failure—where markets fail to produce efficient results. Strength of microeconomics: Simple structure, Strong connection with real-world decisions In summary: Microeconomics = Supply + Demand + Market interactions

Industry-Level Study: Industrial organization explores: Monopoly, Cartels, Different competitive behaviors, How goods/services are sold in various market structures

Macroeconomics: Macroeconomics studies the **economy as a whole** on a bigger scale. It can be applied to: Town, City, Province, National level. **Macroeconomic Factors:** Large-scale patterns or conditions affecting entire economies.

Key Topics in Macroeconomics

1. **Inflation:** Continuous increase in the average prices of goods and services over time.
2. **Economic Growth Rate:** Percentage change in a country's total output (goods + services) over a period.
3. **Price Level:** Overall existing price variation in the economy.
4. **National Income:** Total money earned within a country.
5. **Unemployment Level:** Percentage of people in the labor force who are unemployed.

Scarcity of Resources: Scarcity = **unlimited wants + limited resources**. Examples of scarce resources: Time, Money, Natural resources Anything that is **finite** is considered scarce.

Why Scarcity Matters: Scarcity sets the **value** of goods/services. Scarce resources → Higher demand → Higher price

Resource Allocation: When resources are limited, they must be allocated: Carefully, Wisely (judiciously), Using cost-effective, technically feasible, and long-lasting alternatives

1. **Adam Smith's Theory of Capitalism** — Adam Smith (1723–1790) is known as the Father of Modern Economics. In his famous book “The Wealth of Nations” (1776), he explained the idea of Capitalism.

Main Idea: People work for their own profit, but this unintentionally helps society. This positive effect is called the “Invisible Hand” — meaning people’s self-interest benefits everyone.

Key Features of Capitalism: Private ownership of businesses and property, Freedom of choice for consumers and producers, Competition in markets, Limited government control

Advantages: Encourages innovation, hard work, and economic growth, Gives freedom to produce, sell, and buy anything Improves quality and efficiency because of competition

Bad Consequences: Creates income inequality (rich vs poor). Can cause worker exploitation and unemployment. May lead to greed, monopolies, and less care for social welfare

Conclusion: Capitalism helps economies grow, but it must be balanced with government control to protect people’s welfare.

2. **Karl Marx's Theory of Socialism** — Karl Marx, a German economist and philosopher, developed the theory of Socialism as a reaction to the problems caused by Capitalism. He believed Capitalism creates huge inequality between rich and poor.

Class Division According to Marx

- **Bourgeoisie (Capitalists)** Own factories, land, and means of production
- **Proletariat (Workers)** Sell their labor for wages

Key Idea: Capitalists exploit workers by taking most of the profit from their labor. This profit taken by the capitalist is called surplus value. This exploitation creates class conflict and economic inequality.

Marx's Prediction: Workers (Proletariat) will one day unite and revolt against Capitalists. After the revolution: All means of production (factories, land, resources) will be owned collectively by society, not by individuals.

Aim of Socialism: Create a classless, equal society, Everyone works for the common good. Wealth is shared fairly. End exploitation and ensure social welfare

In Short: Marx's Socialism aims to end worker exploitation, ensure equality, and replace private ownership with **collective ownership**.

3. **Keynes' Mixed Economy** — John Maynard Keynes argued that pure capitalism and pure socialism both have problems. So he proposed a Mixed Economy, where both government and private sector share control.

How a Mixed Economy Works

- **Private sector:** Produces goods and services for **profit**
- **Government sector:** Handles important areas like: Education, Health, Infrastructure

Government Role During Economic Problems: Government can intervene by: Increasing spending, Changing taxes. This creates jobs, improves stability, and supports the economy during downturns.

In Short: Keynes' Mixed Economy = Balance between: Market freedom (capitalism), Government control (socialism). Purpose: Ensure growth, stability, and full employment.

Theory of Demand and Supply

Introduction: Demand and supply is one of the most important concepts in economics. The law of demand and supply explains the relationship between buyers and sellers of a product. It shows how the price of a commodity affects: Buyers' willingness to buy, Sellers' willingness to sell

Basic Idea: When price increases → sellers want to supply more, but buyers demand less. When price decreases → buyers demand more, but sellers supply less.

This theory is based on two laws: **Law of Demand, Law of Supply**. These laws together decide: The **actual market price**, The **quantity of goods** available in the market

Demand & Supply – Introduction: This law is the foundation of most economic principles. In the real market, the equilibrium price is decided by: Quantity people want to buy. Quantity people are willing to sell. Many factors can change demand and supply. In a **perfectly competitive market:** Buyers and sellers act based on self-interest. Consumer's goal → maximize satisfaction. Firm's goal → maximize profits. In equilibrium, both goals match.

What is Demand? Demand means how much of a product a customer wants to buy at a **given price** during a **specific time**.

Demand has two parts: **Willingness to buy** – desire for the product. **Ability to pay** – having money to buy it

To create demand, these three are essential: Desire for the product, Means (money) to buy, Willingness to use money to buy

Law of Demand: When price rises, quantity demanded falls. When price falls, quantity demanded rises. Demand curve shows an inverse relationship between price and quantity demanded (downward sloping).

Determinants of Demand (Factors Affecting Demand)

1. **Price of the commodity:** If price increases → demand decreases. If price decreases → demand increases
2. **Price of related commodities**
 - **Complementary goods** (used together): Example: shoe & socks, phone & cover. If price of one increases → demand for the other falls
 - **Substitute goods** (alternatives): Example: soap & bodywash, notepad & notebook. If price of one rises → demand for the substitute increases
3. **Income of consumers** Higher income → more demand. Lower income → less demand
4. **Tastes and preferences:** Trendy products have more demand. Outdated products have less demand

Type of Demand	Meaning
Price Demand	Quantity demanded at a given price

Cross Demand	Demand of a product changes due to price of related products
Income Demand	Quantity demanded at a particular income level
Joint Demand	Two goods are demanded together (e.g., tea & sugar)
Composite Demand	Goods with multiple uses; demand for one reduces supply of another
Derived Demand	Demand for goods used to produce other goods (e.g., wood for furniture)

What is Supply? Supply means the quantity of a good or service that producers are willing to sell at different prices during a specific time. Supply depends on: Willingness to supply, Ability to supply (availability of the product)

Law of Supply: When price increases, the quantity supplied increases. When price decreases, the quantity supplied decreases.

Reason: higher prices → higher profit. The supply curve is upward sloping, showing a direct relationship between price and quantity supplied.

Demand	Supply
Desire and ability of buyers to purchase goods	Quantity of goods firms are ready to sell
Downward-sloping curve	Upward-sloping curve
Inverse relationship with price	Direct relationship with price
Refers to consumers	Refers to firms
Determinants: tastes, number of buyers, expectations, income, price of related goods	Determinants: resource prices, number of firms, technology, taxes, subsidies

Equilibrium Point: Equilibrium is the price where demand = supply. At this point, both buyers and sellers are satisfied. It is also called: Market-clearing price, Market equilibrium

In the table, at price **6**, quantity demanded = 30 and quantity supplied = 30. This is the equilibrium point.

Why Equilibrium Matters: At equilibrium: market is stable. In disequilibrium:

Surplus or shortage happens, Prices and quantities keep changing, It affects firms, markets, commodities, and the whole economy

Price	Quantity Demanded	Quantity Supplied
10	10	50
8	20	40
6	30	30
4	40	20
2	50	10

MARKET STRUCTURE

What is a Market? A **market** is the place or system where **buyers and sellers interact** to exchange: Goods, Services, Resources

In a market: **Buyers** want to purchase goods or services at a price they are willing to pay. **Sellers** want to sell goods or services at a price they are willing to accept. Markets can be: **Physical** (bazaars, shops), **Online** (e-commerce websites), **Financial** (stock markets)

Markets may operate at: Local level, National level, Global level

The **forces of demand and supply** determine prices of goods and services in a market.

FORMS OF MARKET STRUCTURE: There are **four main types** of market structures:

- **PERFECT COMPETITION MARKET Key Features:** Large number of buyers and sellers, Homogeneous products (all products are identical), Perfect information (buyers & sellers know everything about prices and products), Low barriers to entry and exit (easy to start or close a business), No single buyer or seller can control the market price, Prices are determined purely by demand and supply. **Examples:** Agricultural products, Stock markets, Foreign exchange (forex) markets
- **MONOPOLISTIC COMPETITION MARKET Key Features:** Large number of buyers and sellers. Differentiated products (products are similar but slightly different). Firms differentiate their products through: Advertising, Packaging, Branding. Every seller has some control over price due to product differentiation. **Examples:** Fast food restaurants, Clothing retailers, Toothpaste brands
- **OLIGOPOLY MARKET Key Features:** A **small number of large firms** dominate the market. **High barriers to entry** (difficult for new firms to enter). Firms engage in **strategic behavior**. Price fixing. Collusion (secret cooperation). Firms are **interdependent**. Each firm's actions affect the others. **Examples:** Automobile manufacturing, Telecommunications, Airlines
- **MONOPOLY MARKET Key Features:** Only **one seller or producer** of a good or service, **Extremely high barriers to entry**, The monopolist has **complete control over market price**, Monopoly may arise due to: Government regulation, Patents, Economies of scale (large-scale production reduces cost). **Examples:** Water utilities, Electricity utilities, Certain pharmaceutical companies, Internet service providers in some regions

Financial Statements

Financial statements are reports that show how well a business is doing financially. They help us understand whether the business is making profit, managing its money properly, and staying financially stable.

Types of Financial Statements: There are **four main financial statements**:

- **Income Statement (Profit and Loss Statement):** This statement shows how much money the business earned (revenue) and how much it spent (expenses) during a period. Formula: $\text{Net Income} = \text{Revenues} - \text{Expenses}$. Purpose: Shows whether the company made a profit or loss.
- **Balance Sheet:** This statement shows at a specific moment: What the company owns → *Assets*. What the company owes → *Liabilities*. What belongs to the owners → *Equity*. Formula: $\text{Assets} = \text{Liabilities} + \text{Equity}$. Purpose: Gives a snapshot of the company's financial position at one point in time.
- **Statement of Cash Flows:** This statement shows **how cash enters and leaves the business** during a period. It is divided into three parts: **Purpose:** Shows whether the company has **enough cash** to run daily operations and grow in the future.
 - **Operating Activities:** Cash from normal business work (selling products/services).
 - **Investing Activities:** Cash used to buy or sell long-term assets (like equipment or property).
 - **Financing Activities:** Cash from loans, investments, or paying dividends.
- **Statement of Changes in Equity:** This shows how the **owner's equity** changed over time because of: Profits, Losses, Additional investments, Withdrawals by the owner **Purpose:** Shows how ownership value changes due to business performance and owner activities.

Income Statement Example

Item	Amount
Revenue	\$10,000
Expenses	-\$4,000
Net Income	\$6,000

Balance Sheet Example

Item	Amount
Assets	\$8,000
Liabilities	\$2,000
Equity	\$6,000

Why Financial Statements Matter Financial statements are important because: They help check if the business is earning profit or spending too much. They show whether the company has enough cash to pay bills and grow. Public companies must show these statements by law to keep investors informed and transparent.

Lecture Week 3 — Time Value of Money (TVM)

What is the Time Value of Money (TVM)? The Time Value of Money (TVM) is an important financial idea. It means: Money has different value at different times. Money today is worth more than the same amount in the future. This is because money today can earn interest, create returns, or be invested to make more money.

Two Main Principles of TVM

- **Present Value (PV):** PV means the current value of a future amount of money. It tells us how much a future cash flow is worth today when discounted. PV helps calculate the value of future money in today's terms.
- **Future Value (FV):** FV means the value of today's money at some point in the future. It includes the rate of return or interest rate. FV shows how much an investment or savings will grow over time.

Annuity Series: An **annuity** is a series of **equal payments** made at **regular intervals** (monthly, yearly, etc.). Conditions for an annuity: Payments must be **equal**. Time interval must be **regular**. Example: Monthly installments, pension payments, insurance payments.

Money–Time Relationship

1. **The relationship between time and money is dynamic.** Money changes in value over time. Example: What you could buy with \$10 in the past cannot be bought with the same \$10 now.
2. **Purchasing power changes over time.** This happens due to changes in a country's economic conditions:
 - **Value of local currency vs US dollar:** Dollar acts as a standard measure. If currency weakens, purchasing power falls.
 - **Inflation:** Prices of goods and services increase. Manufacturing costs rise → prices rise → money buys less.
 - **Exports vs Imports:** Exports bring dollars into the country. More exports strengthen the economy.
 - **Availability of foreign exchange:** More foreign exchange increases economic stability.
 - **Reserves in Current Account:** High reserves create a **positive impact** on the economy.
3. **Because money is dynamic,** money today is worth more than money in the future. Its earning capacity reduces over time.
4. **Money received earlier is better than money received later.** The sooner you earn or receive money, the more benefits you get. This is the core idea of TVM.
5. **Money creates more money.** Money grows when invested or saved.
6. **Money grows mainly through interest.**
7. **Interest is considered a sin (usury) in Islam and Christianity.**
8. **Pakistan's banking system has shifted to Islamic Banking (Modaraba system).** Modaraba involves profit-sharing instead of charging interest.
9. **TVM explains why interest is earned or paid.**
10. **Money deposited in the bank is not kept idle.** Banks invest the money in businesses. Business earns profit → Bank gives a share of profit to the depositor. This share is called an **interest rate**.
11. **Interest compensates for the Time Value of Money.** Depositors accept delaying their spending only if they expect a good return in the future.
12. **Economics helps calculate the value of money:** In the future (forward in time) In the past (retrospective or backward in time)
13. **Present value is calculated using a discount rate.** Discounting is used when calculating present value (backward). Interest rate or compound interest is used when calculating future value (forward).

14. Using tables and mathematical formulas, we can calculate PV and FV: For investments over long periods—even up to 100 years

Benefit Cost Ratio (BCR)

What is Benefit Cost Ratio (B/C or BCR or BCA)? The Benefit Cost Ratio (B/C), also called Benefit Cost Analysis (BCA), is a financial technique used to evaluate a project or compare two alternatives. It helps determine whether the benefits of a project are greater than its costs. B/C ratio is also used to calculate: Present Value (PV), Annual Cost, Rate of Return (ROR). The main goal is to compare two alternatives using the benefit/cost ratio.

Decision Rule: $B/C > 1 \rightarrow$ Project is desirable. The higher the benefits, the better the project.

General Formula: $B/C = AWB / AWC$ (AWB = Annual Worth Benefit, AWC = Annual Worth Cost)

Methods to Calculate B/C Ratio:

- **Conventional B/C Method:** The **Conventional Method** came earlier and has a limitation: **Dis-benefits (DB)** cannot be calculated in this method.
- **Modified Method:** The Modified Method was developed later. It allows the inclusion of **Dis-benefits (DB)**, which the conventional method does not.

Problem 2 (a) Benefits and Disbenefits of Building a Dam

Benefits: Water storage, Hydroelectric power, Irrigation support, Flood control, Economic development, Job creation

Dis-benefits: High construction cost, Environmental damage, Displacement of people, Loss of wildlife habitats, Maintenance cost, Risk of dam failure

Income Statement

What is an Income Statement? The Income Statement (also called Profit and Loss Statement) shows how much money a business earns and spends over a specific period (month, quarter, or year). It helps us understand whether the business is making a profit or facing a loss.

What the Income Statement Shows: The Income Statement works like a **report card** for a business. It answers: How much money did the business make? How much money did it spend? Did it end with a **profit** or a **loss**?

Parts of the Income Statement: An Income Statement includes several sections that help calculate the **final profit (Net Income)**.

1. **Revenue (Sales):** Total money earned by selling products or services.
2. **Cost of Goods Sold (COGS):** The cost of making or buying the goods that were sold. Example: For a furniture business, COGS includes wood, tools, and labor.
3. **Gross Profit:** Money left after subtracting COGS from Revenue. It shows how much the business earns **before** paying other expenses. Formula: $\text{Gross Profit} = \text{Revenue} - \text{Cost of Goods Sold}$
4. **Operating Expenses:** Costs needed to run the business, such as: Rent, Salaries, Utilities, Office expenses
5. **Operating Income:** Profit from regular business operations. Calculated after subtracting Operating Expenses from Gross Profit. $\text{Operating Income} = \text{Gross Profit} - \text{Operating Expenses}$
6. **Other Income and Expenses:** Things not part of daily business operations: Interest income, Interest expenses, Gains or losses outside normal activities
7. **Income Before Tax:** Profit remaining before paying tax.
8. **Income Tax Expense:** The amount of tax the business owes.
9. **Net Income (Net Profit):** Final profit after subtracting all expenses including taxes. $\text{Net Income} = \text{Income Before Tax} - \text{Income Tax Expense}$

Importance of the Income Statement: Shows whether the business is **profitable** or **losing money**. Helps managers and investors find: Which areas make money. Where expenses can be reduced. Supports decision-making such as: Expanding operations, Cutting costs, Improving business strategies

Break-Even Analysis & Motor Cost Comparison

Break-Even Analysis (B–E Analysis): Break-even analysis compares Cost vs Income to find the point where a business makes: No Profit, No Loss. This point is called **Break-Even Point (BEP)**.

Components of Cost

A) Fixed Cost (FC): These costs do **not change** with the number of units produced: Capital cost, Cost of land, Taxes (GST), Salaries. Rent, Utility bills, O&M cost, Machinery cost

B) Variable Cost (VC): These costs **change** depending on production: Production cost, Labour, Materials, Machinery usage, Technology, Utility bills, O&M, Salaries (if production-dependent)

Students also know: Factors of production: Land, Labour, Capital. Laws: Increasing Returns, Decreasing Returns, Constant Returns

Income / Revenue: $\text{Income} = \text{Total revenue received from selling products}$. Also called: Receipts, Return on Investment (ROI)

Break-Even Condition: Break-Even occurs when: **Cost = Income, TR = TC** (No profit, No loss)